

Data Flow Diagram

Date	29 June 2026
Team ID	
Project Name	Rising Waters: Smart AI Flood Prediction System
Maximum Marks	2 Marks

Data Flow Diagram

Create a Data Flow Diagram (DFD) to show how data moves through your system - between external entities, processes, and data stores. Use the symbol legend below as a guide.

DFD Symbol Legend

Symbol	Name	Description
Oval / Rounded shape	External Entity	• Represents users or organizations that interact with the flood prediction system. • Provides weather and rainfall data required for prediction. • Receives flood prediction results and warning notifications. • Examples include Meteorologists, Disaster Management Officers, Government Authorities, and Emergency Response Teams.
Rectangle with numbered header	Process	• Accepts input weather parameters from users. • Performs data preprocessing, feature scaling, and validation. • Applies machine learning algorithms (Decision Tree, Random Forest, KNN, XGBoost) to predict flood risk. • Generates and displays the final flood prediction result through the Flask web application.
Rectangle (solid fill, no number)	Data Store	• Stores historical weather and flood datasets used for model training. • Maintains the trained machine learning model (XGBoost) for prediction. • Saves prediction history and user input records for future analysis. • Provides data whenever the prediction engine requests it during execution.

Labeled arrow	Data Flow	• Shows the movement of weather and rainfall data from users to the prediction system. • Transfers processed data between preprocessing, model prediction, and result generation modules. • Carries prediction results and flood alerts back to users and authorities. • Ensures smooth communication between external entities, processes, and data stores throughout the system).
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Example

The diagram below illustrates how the symbols above come together to form a complete data flow diagram.



